

# Financial Risk & Fraud Analysis

Power Query • MySQL • SQL • Power BI • DAX

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# Project Overview

Objective, data source, and analytical approach

## OBJECTIVE

- Analyze financial transactions and loan portfolios.
- Identify meaningful risk indicators and unusual patterns.
- Build an interactive dashboard that supports decision-making.

## DATASET

- Synthetic Finance Fraud & Loans dataset.
- 11 related tables covering customers, accounts, transactions, loans, branches, and reference data.
- Data quality issues were intentionally included for cleaning and analysis.

## WORKFLOW

Power Query  
Cleaning

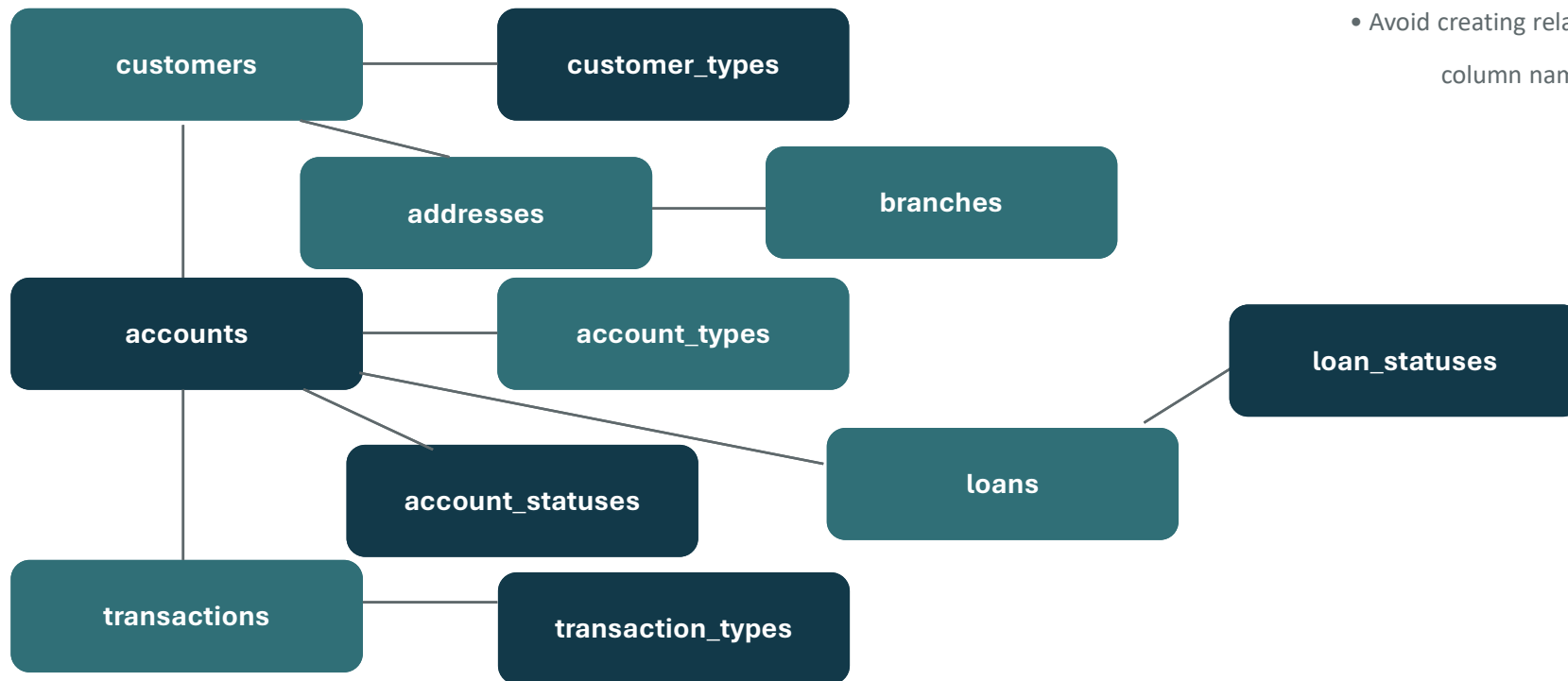
MySQL  
SQL Analysis

Power BI  
Modeling + DAX

Dashboard  
Insights

# Data Model & ERD

Key relationships used across the financial dataset



## Modeling principle

- Primary keys are kept unique.
- Foreign keys connect related entities.
- Avoid creating relationships only because column names look similar.

# Data Cleaning – Power Query

Cleaning decisions made before analysis

## Missing values

Replaced appropriate text blanks with “Unknown”; kept missing dates such as DateOfBirth as null when inventing a value would be misleading.

## Duplicates

Removed exact duplicate rows across all columns rather than deleting rows based on an ID alone.

## Text quality

Used Trim and Clean to remove extra spaces and hidden/non-printing characters.

## Date fields

Reviewed date quality and handled missing OpeningDate with a simple mode-based approach where chosen.

## Outlier review

Used Average, Standard Deviation, and relative dispersion to understand unusually spread values.

# SQL Analysis – MySQL

Queries used to turn cleaned data into business findings

**JOIN**

Connect related tables to enrich the analysis.

**CASE WHEN**

Create business categories such as balance levels.

**GROUP BY**

Compare metrics by customer, transaction, or loan category.

**VIEW**

Save reusable analytical logic.

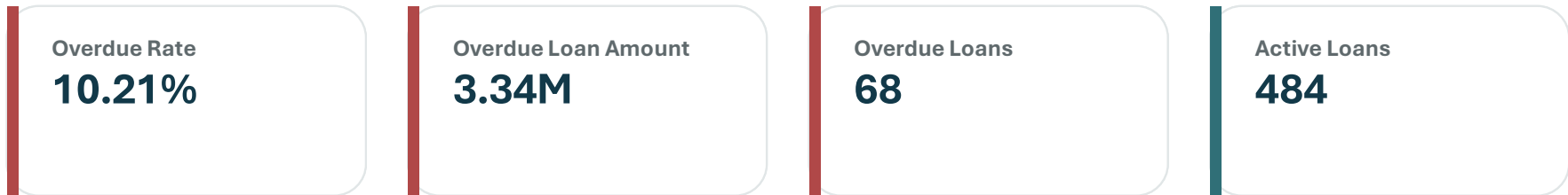
**Stored Procedure**

Create a reusable parameterized query for customer/account analysis.

**Example:**  
**Overdue loans by customer type**

# Key SQL Findings

The analysis moved from descriptive counts to risk-focused indicators



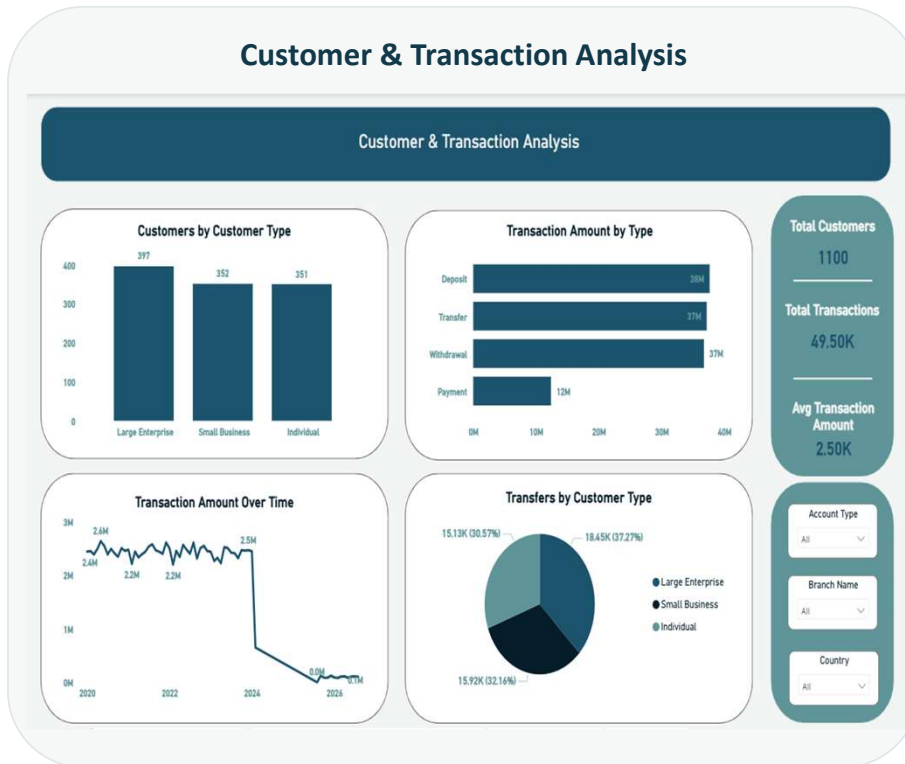
## INSIGHTS

- Large Enterprise had the highest number of overdue loans among the customer types analyzed.
- Overdue loans represent a meaningful credit-risk indicator compared with simply looking at high account balances.
- Average transaction amount was about 2,498.55, with standard deviation about 1,443.54, indicating noticeable variability.

*Note: KPI values can change when dashboard date filters are applied.*

# Power BI Dashboard

Three-page story: overview → customer behavior → risk



# Dashboard Pages & Visual Logic

Each page answers a different business question

## 01 Executive Overview

### What is the overall financial picture?

KPI cards • transaction volume • loan status • outlier scatter

## 02 Customer & Transaction Analysis

### Who are the customers and how do they transact?

Customer type • transaction amount by type • trend • transfers by customer type

## 03 Risk Analysis

### Where are the main loan-risk signals?

Overdue rate • overdue amount • overdue loans • status • customer/account type • loan scatter



# Key Insights & Business Interpretation

What the dashboard tells us

## Credit risk

10.3% of loans are overdue in the Power BI view, making overdue exposure a key risk metric.

## Customer segment

Large Enterprise shows the highest overdue-loan count among the customer types in the analysis.

## Transaction behavior

Deposit, Transfer, and Withdrawal carry much larger transaction volumes/amounts than Payment.

## Data variability

Transaction amount has meaningful dispersion; this supports continued monitoring of unusual transaction values.

## Time filtering

Date slicers materially change totals, so KPI interpretation should always consider the selected period.

## Recommendations & Conclusion

- Prioritize monitoring of overdue loans and overdue exposure.
- Segment risk monitoring by customer type and account type.
- Investigate unusually high transaction amounts rather than treating high balances alone as risk.
- Use date filters to compare risk trends across periods.
- Maintain the Power Query → MySQL → Power BI workflow for repeatable analysis.

### Final Takeaway

A clean data pipeline + focused SQL analysis  
+ interactive  
Power BI storytelling turns raw  
financial data into actionable risk insights.